

Architectural Justification

Architecture Selection

We chose Layered Architecture for the Faculty Research Grant & Publication Tracker System.

Reason 1 - Separation of Responsibilities

The system contains distinct functions for grant management, publication management, approval workflows, and grant analytics. Layered Architecture separates the user interface, business logic, and data storage, making these responsibilities easier to organize, understand, and maintain.

Reason 2 - Suitable for a Centralized Academic Portal

Faculty Researchers, Co-authors, and the Research Dean access a common portal and shared data. Requests can move from the Presentation Layer to the Business Layer and then to the Data Layer in a clear and controlled manner, which suits the centralized nature of the system.

Security Advantage

Validation, authorization, and audit controls can be enforced in the Business Layer before changes reach the Data Layer. This helps protect sensitive grant and publication operations and supports secure handling of financial approvals and status modifications.

Performance Benefit

Separating business logic from data access allows database queries and frequently accessed grant, publication, and analytics information to be optimized independently. This supports responsive access under normal and peak usage without adding the operational complexity of a microservices architecture.